Corrections and Clarifications

FFGFT / T0 Time-Mass Duality

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Contents

Preliminary remark

This document is the compact, running correction register of the FFGFT corpus. It supersedes the previous extended edition, which remains available unchanged as **Doc. 190-Archive** (`190_T0_Korrekturen_Archiv_De/En`) and contains, for every entry, the full justification, the faulty and corrected expressions verbatim, and all addenda. The table below lists *all* entries (C/K, P, R) with the affected documents and the subject in short form. The register remains append-only: source documents are not revised (cf. R50); new entries are appended here, and their elaboration will henceforth be written directly in this document.

.1 Register table

C/K = correction, P = clarification, R = revision/register entry. Full texts: Doc. 190-Archive.

No.	Affects	Subject (short form; details: Archive)
C1	049 (HTML)	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ *[Addendum 28 July 2026, cf. R66: reason for the correction —* $16\pi^3 = (4\pi)\cdot(4\pi^2) = S^2 \cdot 2S^3$ *is the sphere count of the compact* T^4 *boundary geometry (Doc. 310); the earlier version counted one 2-sphere too many]*
C2	116 (tab.)	$r_\tau = 25/9$
C3	018 (Explorer)	$a_e = 2\pi^2/(f \cdot k)$
C4	267 (DE+EN), 268 (DE)	**Factor 3 = three observable space dimensions:** the observer measures $k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$ (the fourth torus direction is time unfolding).
C5	275 (script v3)	**All three fixed in v4** (11 June 2026): T^4 as kNN in genuine 4D coordinates with dimension-matched 4D nulls (result: $1{,}329 \pm 0{,}025$ *inside* the band $[0{,}690,\ 2{,}406]$); HLV ensemble with $\sigma = 0{,}05$ ($0{,}710 \pm 0{,}008$, in the band); the shell detect ...
P1	116 / 158	both correct; $0{,}001\,\%$ for external use
P2	several	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
P3	173–176	two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)
P4	180–182	L_0 = geom. fundamental length
P5	several	mass structure from ξ; $\sim 1\,\%$ agreement
P6	116 / 158	Koide only with numerical values (non-closure, Doc. 193)
P7	005, 008, 019, 086, 189, 202	scale structure from the recursion $\mathscr{R}$
P8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	geometric scale adjustment / fractal correction
P9	005, 006, 042, 046	columns “scaling class” + “SM correspondence”; quarks via $m = r\xi^p v$
P10	009	$K = R_{pe}/(4/3)^7$: an empirical anchor; factor $1{,}314$ defined implicitly, not derived
P11	025, 003, 133	correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0{,}0002\,\%$; Doc. 003 and Doc. 133 use different models
P12	022, 230	two different observables: cumulative (Doc. 022) vs. elementary $\xi/(2\pi)$ (Doc. 230); 10^{-5} unreachable at any finite N from Doc. 022
P13	013, 025, 061	$\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} = m_e c^2/(\frac{16}{9}\ln 2 \cdot \text{eV}) \approx 414\,684$; table in Doc. 013 (archived) reconstructed

No.	Affects	Subject (short form; details: Archive)
P14a	041	fractal path lengthening (geometric path stretching); energy loss only an artifact when the path lengthening is omitted; path-length dependence remains (Doc. 026/149/182); the same path lengthening carries the CMB treatment in Doc. 268 (formerly tracked separately as P19)
P15	061 / 041	marked as a ΛCDM comparison quantity; the static universe has no cosmological z; CMB by stationary thermalization
P16	026 / 064 / 181	H_0 is emergent (decompactified), not fundamental; implemented in Doc. 263, Docs. 026/064/181 noted as outdated
P17	028	the DE/DM relation is circular (gives 2.5 for any ξ); MOND partially derived ($\xi^{1/4}$ genuine, K_M free); CMB relations unaffected; noted
P18	267	the two pictures are *not* distinguishable through the observed ratios; the decision is theoretical, not empirical (conceptual placement)
P20	267 / 268	the *form* is fixed: a dimensionless ξ ratio (R_H/ℓ_P).
P29	268	*Only partially answered.* The factor 3 from the 3D projection ($3 \cdot \{1, 6, 14, 26\}$) is plausible, but the sequence $\{1, 6, 14, 26\}$ is *read off the data* (Doc. 268, step 4), not produced forward.
P30	268	the *naive* spherical Bessel sum $C_\ell = \sum_n N_{\mathrm{BE}} \lvert j_\ell(k_{\mathrm{obs}} D_A)\rvert^2$ does *not* reproduce $\{3, 18, 42, 78\}$ – it washes out to a quasi-continuum dominated by low modes.
P31	268	*Leading behaviour derived forward.* Forward, T^4 gives the harmonic backbone $1 : 2 : 3 : 4 : 5$ (symmetric resonance $k^2 = 3n^2$).
P32	corpus-wide	follows from P20: H_0 is an *imported* external calibration, not derived.
P33	009 / 042 / 181	the *form* of ξ is forward: the harmonic $4/3$ (perfect fourth, Doc. 042/159) gives the leading digit, 3 (space dimensions) and $2^2{=}4$ (spacetime) are read off the geometry.
P34	181 / 013	*3+1 is empirical input* – no theory derives the dimension count; the earlier "follows necessarily" claim (Doc. 181) is overstated.
P35	corpus-wide (esp. 182, 250s, scale tables)	ξ^N **triviality:** "$X = \xi^N$" is by itself *vacuous*, since any magnitude can be written as a ξ power – an exponent alone is not a prediction (cf. P10, P20, P33: form forward, number input).
P36	182 / cosmol. sector	R_H **interpretation refined:** $R_H = c/H_0$ is the *Hubble length* – the local expansion rate expressed as a distance – *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM).
P37	009, 188, 271, 272, 275	**Three levels clearly separated:** **(1) Microscopic – input:** $\xi = 4/30000$ (fractal dimension).
P38	275	**Status clarification:** (a) $1/\varphi$ is a *pass-through point*, not a fixed point – the only fixed point of $r \mapsto r(1-\xi)$ is 0.
P39	182, 267, 277, corpus-wide (H_0/R_H)	H_0**/**R_H **status refined (shared reference point, not an FFGFT gap):** the relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6{,}49 \times 10^{64}$) is derived from first principles by *no* framework – why the universe is $\sim 10^{60}$ Plan …
C6	004, 025, 030, 039, 041, 053, 061, 063, 068, 081	**Redshift is achromatic.** The fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor: $1 + z = e^{\xi x}$, independent of λ.
P40	005, 006, 011, 012, 133, 275	**Only ratios are exact.** Absolute values carry the fractal/SI correction, absorbed into the bridge constants v (masses $m = r\xi^p v$), $E_0 = 7{,}398$ MeV (α) and $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; `calc_De.py`); in ratios they cancel ($m_\mu/m_e = 206{,}768$ …

No.	Affects	Subject (short form; details: Archive)
P41	267, 280, 221 (cosmological sector)	**Checked and not booked (degeneracy risk at the shared scale).** ΛCDM signal $\Delta v = c\,\dot{z}\,\Delta t/(1+z)$ over 25 years: $+6$ cm/s (z=0.5), sign change at $z_0 = 1.92$, -14 cm/s (z=4), -20 cm/s (z=5); ELT/ANDES order $\sigma \sim 2$ cm/s.
P42	282, 292 (lepton sector)	**One declared reference point: the ratio m_μ/m_e.** The framework fundamentally keeps the purely theoretical derivation from ξ; the lepton sector receives – analogous to P39 (H_0/R_H as shared empirical reference) – exactly *one* declared reference point ...
P43	006, 011, 292 (`calc_De.py`)	**No rework; recorded as a note.** Verified: `calc_De.py` does *not* use $K_{\text{frak}} = 1 - 100\xi$ explicitly.
R41	284 (cf. 270, 249)	**The loss is not the SI conversion.** Per Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T} \cdot m = 1$) is a *mode-preserving embedding* (lossless); compactness survives as the discrete spectrum and as physical observables (T_{rev} ...
R42	270 (cf. 248)	**Not an open geometric question – it is the measurement.** In the compact view the four circles are symmetric (no space/-time distinction); time is not selected by the geometry but is *relational* – the symmetry-breaking *is* the measurement (the Type I decompact ...
R43	285, 283	**Corrected to containment.** $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron, `recompute_reconcile.py`) and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$), so HLV ⊃ FFGFT at the group/dimension level – the models are reconcilable, not inco ...
R44	272, 190 (P35), 285	**Sharpened (no contradiction).** φ acquires meaning *not* inside FFGFT but as the derived eigenvalue of HLV's 5-fold / perp sector (golden inflation $\varphi^2 = \varphi + 1$) – a marker of the boundary HLV∖FFGFT.
R45	285, 282–284	**Sharpened.** A quasicrystal has two spectra: the *structure / diffraction* spectrum is pure point (sharp Bragg peaks, $\approx 21\times$ above a random null, `quasicrystal_information.py`) – that is the information and exactly HLV's substrate-distinguishabilit ...
R46	296 (cf. 025, 026, 063, 156)	**Open check-point (not yet processed).** This intermediate-scale formulation potentially runs *counter* to the already established corpus line, which treats dark matter/energy not as an open signature but as a *derived consequence*: Doc. 025 reads dark matter as a ...
R47	285, 297, 298 (cf. 286, 284, 295)	**In the time sector: containment, not enlargement.** In unified notation – both $D6+$time, Marcel as $D6+$spiral-time, FFGFT in Hilbert space as $D6+$recursion-time – the spiral-time window is not additive but *internal*: a bounded section (window) of the unrolled re ...
R48	304 (cf. 295, 301, 303)	**Note withdrawn, document made precise (not repaired).** The overstatement lay solely in the *mail* and was withdrawn without reservation; the document was written throughout from the FFGFT reading and already booked the bridge as restricted.
R49	304, 305 (cf. 295, 303, 272)	**Vocabulary cited, marked as analysis, abstract desymbolised — notation not renamed.** A consolidated first-use footnote attributes U1/U2/U3 *and* M_τ/M_B to Krüger (2026) with the primary source (`doi:10.5281/zenodo.21302278`) and documented lineage (Me ...
R50	025, 063, 061, 064, 078 (superseded by 306; cf. 003, 010, 295, 298, 302)	**Justification replaced, conclusion unchanged.** The conclusion (no singular beginning; a finite core with a minimal scale L_0 set by ξ) stands.

No.	Affects	Subject (short form; details: Archive)
R51	049 (cf. 095, 306, 267, 302; notation)	**A notational relic, not a substantive contradiction — and nonetheless no longer authoritative.** *(a) The resolution:* in the original T0 view, T_0 never denoted the instant zero but the *smallest possible time interval*.
R52	101, 165 (numerical values; cf. 180, 182, 013, 250, 290)	**Values corrected; the main relation is unaffected.** *(i) Wrong* L_0: correct is $L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \times 10^{-35}$ m = $\boxed{2.155 \times 10^{-39}\ \text{m}}$ — as in Doc. 180 (the dedicated derivation), Docs. 013, 181, 182, 187 ...
R53	267 §518 (table row *source of the scale*; cf. 182, 165, P20/P35/P36)	**Corrected reading — and the symmetry becomes sharper, not weaker.** *(a)* On the FFGFT side, the table row *source of the scale* correctly reads not "from ξ" but **"set / calibrated to H_0"**.
R54	292, 306 (cf. 307)	**(a) N_0 is not a base unit.** $N_0 \approx 38.6$ is a *ladder-internal*, generation-counting factor ($N_g = g N_0$, Dok. 292); its value is *open* (P35: $100/e$, $100/\varphi^2$, 39 do not hit it reliably).
R55	292 §Cross-ladder coupling (cf. R54, P35, P40/P41)	**The law stands; its evidential breadth is one entry, not three.** *(i) The third entry is forced arithmetically.* From $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ a uniformly multiplicative correction satisfies $K^{p_3} = K^{p_1} K^{p_2}$, hence $p_3 = p_1 + p_2$ *identicall ...*
R56	188 §Special role of $n = 3$ (cf. 189 §Derivation hierarchy, 248, 230/231/232, R55)	**The three elements are hereby booked uniformly as foundational assumptions.** *(i) The declared framework.* FFGFT posits three structural elements: the fundamental relation $\tilde{T} \cdot m = 1$ (Doc. 189, level 1), the topology T^4 (Doc. 248), and the order Z_3 ...
R57	A015 (cf. 020, 030, A-Series)	**Demand dissolved — ξ is a fundamental parameter.** A proof is neither possible nor necessary; the demand was wrongly posed.
R58	A142, A140 (cf. 127, 163)	**Gravitational quantisation not required.** $G = \xi^2/(4m_{\text{char}})$ is intrinsic to the torus geometry; a separate quantum gravity does not arise, because the geometry T^4/Z_3 is pre-ordered and not dynamical in the same sense as quantum-field-theoretic fields.
R59	A130 (cf. 011, 043, 101; P40)	**Structurally explained [S].** Two causes: (i) SI values are scheme-dependent (1-loop, $\overline{\text{MS}}$), i.e. the EFT precision values are not theory-independent raw data; (ii) fractal corrections in v are not cleanly separable from the loop structure of the SM ...
R60	A160, A165 (cf. 020, 022, 023, 230; P35)	**Both factors now geometrically derived [B].** (i) 2π is the circumference of the unit circle — the natural conversion between the torus winding phase and the measurement angle; $\xi/(2\pi)$ is thus the phase correction per measurement.
R61	174, 175 (cf. A142, 147, 162; R57, P35)	**Data-driven value selection — not a second parameter; withdrawn.** *(i) The genesis:* ξ_{Higgs} was selected *because* it lands in the measured window; the two-spaces doctrine is rationalisation after the selection, not independent justification.
R62	005 §Baryon worked example (proton) (cf. R50, P35)	**The anchor of the formula line is $\pi^2/2$, not m_μ.** *(i) Diagnosis.* The missing factor identifies the error as a single symbol: $(\pi^2/2)/m_\mu = 46.705$ hits the missing factor to $0.07\,\%$.

No.	Affects	Subject (short form; details: Archive)
R63	253 §ξ resonance heuristic (code: `rsa/factorization_benchmark_library.py`; cf. 024, 075, 076, 176; R57, P35)	**ξ is inert in the "optimal" range; the optimisation switched the filter off.** *(i) Inertness.* The code threshold is $1/1000$.
P44	253, 190 (R63); cf. A135, P35	**Pre-registered, tested, and refuted for this design; the window width is a property of the measurement setup.** *(i) Design.* Shor phase estimation, 12 counting qubits (phase resolution $2^{-12} = 2.44 \times 10^{-4}$), 8 shots, $10\,\%$ fully random outcomes, 300 ...
R64	`2/python/t0_no_fallback_study.py`	**The procedure is right, the separating cut too coarse, and the label wrong.** *(i) What holds.* The counters `t0_success_count` and `fallback_success_count` are incremented at separate places (l. 376 resp. 280/300), `get_statistics()` report ...
R65	024, 075, 076, 147, 176, 253 as well as the	**No procedure distinguishable from Shor exists; the method claim is withdrawn.** *(i) What is withdrawn.* (a) The claim of an independent T0 resp. ξ factorisation procedure.
R66	097 (DE/EN and the standalone versions under	**The $16\pi^3$ count is right — but not for the reason given in Doc. 097.**
C7	the fourteen HTML pages under `rsa/`	**Rebuilt, not annotated: the pages now say what they do.** No correction box was inserted; what was corrected is the computation, the labelling and the method text.
R67	A040 (fractal correction), A270 (bulk exponent 36),	*(i)* With $D_f^{\mathrm{eff}} = 2.973$ (three decimals) the rounding interval leaves $n \in [98.3,\ 102.0]$ open; even $D = 2.973$ exactly gives $n = 100.12$.
R68	279 (H_0 chain), 308 (Λ reading,	*(i) Identification:* the Λ function is carried by the derived quantity $\Lambda^* := 1/R_H^2 = (\pi/2)^2\xi^{20}/\bar{\lambda}_e^2 = 5.218{\times}10^{-53}$ m^{-2} ("closure scale").
R69	308 §a_0 derivation; cf. P20, R68;	*(i)* From $\partial\mathrm{T}^4 = \varnothing$ (Doc. 306) an isotropic torus yields a compact time cycle $\tau_c = 2\pi/H_0 = 91.9$ Gyr [K|P20]; the energy quantum is exactly $\hbar H_0$, identical to the momentum quantum of the spatial cycles.
R70	061 (temperature/CMB), A085 (natural units),	T_0 scale gap
R71	309 (scale anchor, inheritance problem);	sharpening: side-taking in the Hubble tension
R72	A010 (glossary), A080 (units), A265	E_0: bare and corrected
R73	026 (geometric cosmology), 279	H_0 notations, exponent $41/4$
R74	025 (Cosmology), 063 (Cosmic), DE+EN each,	lithium problem: mechanism, not solution
R75	024, 034, 075, 076, 147, 176: ξ name collision and rewrite of the Shor documents	In all affected documents the search parameter is now denoted σ and placed as a sampling grid without physical meaning.
R76	318	Scope of mass derivations
R77	041, 160, 318	Classification of early coupling constant formulas

No.	Affects	Subject (short form; details: Archive)
R78	160, 005	α_s and running coupling
R79	321, 319, 049	Algebraic derivation of the $SU(3)_c$ gauge group
R81	323, 321, 160, 320	Weinberg angle at M_Z from ξ
R82	322, 320	Hilbert-space embedding and spectral theory
R83	319, 321, 323, 322, 320	Status update: closed bridges
R84	324, 321, 009	Casimir derivation of ξ; ξ not a G(6) spectral value
R85	313, 325, 257, 321, 322	Hawking mechanism closed microscopically
R86	Docs. 180, 322, 314, 321, 325	Self-adjointness of $\hat{F}$ fully closed; $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$ **[B]**
R87	Docs. 320, 322	Δm^2_{32}: numerical values corrected, dev. $+16.0\,\%$; candidate analysis added **[K]**

.2 Currently open bridges (as of R85)

- Δm^2_{32} atmospheric ($+16.0\,\%$); mixing term F_5–F_7 — Doc. 320, R87 [S]
- 2-loop corrections to $\alpha_s(M_Z)$ [S]
- m_{Pl} and $\alpha_{\text{em}}(M_Z)$ from ξ (external inputs in R81) [S]
- Quark/hadron sector — Doc. 318, R76
- Greybody factors; back-reaction on orbifold fixed points — R85 [S]
- Forward derivation of the cosmic exponent $41/4$ (P20); the 10^{123} question, $R_H \to \ell_1$ and $\kappa = \Lambda_{\text{obs}} R_H^2$ hang on it
- CMB peak selection: generate $\{1, 6, 14, 26\}$ forward; the $|n|^2{=}30$ case (P29/P31)
- C_ℓ projection: physical source/window function (P30)
- Ω_{DM} quantitatively, model-neutral (P17)
- Spectral dimension $d_s = 1.86$ vs. FFGFT ≈ 2
- Corpus-wide H_0 language cleanup (P34); fundamental rework of Doc. 026 (P16)
- ξ as line width (P44, pre-registered); ι embedding HLV$\subset$FFGFT in the time sector (Doc. 297)

R86 – Self-adjointness of $\hat{F}$ (August 2026): Doc. 327 closes the gap declared as [S] in R82 completely. The finiteness of the scale ladder is guaranteed by $L_0 = \xi \cdot \ell_P$ (Doc. 180 **[K]**) – no infinities in FFGFT, hence $\hat{F}$ is bounded and $\mathscr{D}(\hat{F}) = \mathscr{H}$. The symmetry $\hat{F} = \hat{F}^\dagger$ follows from the $\mathbb{Z}_3$ pairing $(k, -k)$ (the same as in Doc. 325 **[B]**). Deficiency indices $(0, 0)$: exactly one self-adjoint realisation **[B]**. Fractal measure (finite 100-step recursion), $\mathbb{Z}_3$ restriction and all three χ-twist classes preserve self-adjointness **[B]**. $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$ well-defined and truncation-stable (min-max) **[B]**. No remaining cases. Verification script: `327_selbstadjungiertheit_F_verify.py` (21 assertions).
Declared 23 August 2026

R87 – Δm^2_{32}: corrected numerical values and candidate analysis (August 2026): In Doc. 320 the evaluation of $\xi^{9/5}$ was faulty (8.717×10^{-8} instead of 1.059×10^{-7}); the mass formula $m_{\nu_i} = m_e\, \xi^{p_i}$ and the exponents $p_i \in \{9/4,\ 2,\ 9/5\}$ were always correct. Docs. 320 and 322 have been brought to the correct numerical values: $m_{\nu_3} = 54.11$ meV **[K]**, $\Delta m^2_{32} = 2.846 \times 10^{-3}$ eV² **[K]** (deviation $+16.0\,\%$ against NuFIT 5.3), $\sum m_\nu = 64.17$ meV **[K]** (still within the Planck limit). Unaffected: m_{ν_1}, m_{ν_2}, Δm^2_{21} ($+8.3\,\%$) and the fixed-point assignment. Doc. 320 was at the same time extended by a candidate analysis for the remaining deviation: K_{frak} at fixed point

F_7 ($+12.9\,\%$, insufficient), mixing term F_5–F_7 (right sign, complete mass matrix outstanding – the most promising candidate), alternative exponents (unlikely: $p_3^{\mathrm{PDG}} = 1.8081$ lies only $0.45\,\%$ above $9/5$). Δm^2_{32} remains **[S]**. Verification script: `320_dm32_neutrino_verify.py` (10 assertions).

Declared 23 August 2026